

L9: Degree Block Approximation

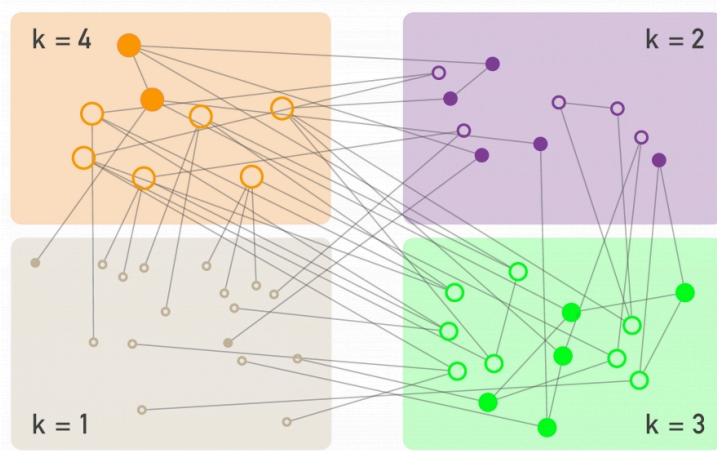


Figure 10.9 from *Network Science* by Albert-László Barabási

To avoid the homogeneous mixing assumption, one option would be to model explicitly the state (e.g., *susceptible*, *infected*, *removed*) of each node in the network, considering the degree of that node. That would result in a large system of differential equations that would only be solvable numerically.

Another approach is to group all nodes with a certain degree k together in the same “**block**”. Then, we can ask questions such as: what is the rate at which nodes of degree k move from the S to the I state? In other words, we will not be able to make specific predictions for individual nodes but will be able to characterize the compartmental dynamics of all nodes that

have a certain degree. This is referred to as the “**degree block approximation**.”

This analytical method can be applied to networks with arbitrary degree distribution (*including power-law networks*). The degrees of neighboring nodes however should be independent. So, even though the degree block approximation is much more general than the homogeneous mixing assumption, it is still not applicable in networks that have strong assortativity or disassortativity, clustering, or community structure.